

Ahan Bhatt

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SUMMARY

Recent M.S. graduate in Computer Science and Engineering with experience building transformer compilers, GPU-accelerated LLM inference systems, adversarial evaluation infrastructure, and trustworthy AI platforms. Strong background in Python, C++, CUDA, Triton, PyTorch, FastAPI, and full-stack ML systems, supported by research experience and publications across LLM evaluation, explainable AI, time-series modeling, and applied machine learning.

EDUCATION

University of California, Santa Cruz [3.82 GPA]
Master of Science in Computer Science and Engineering

Santa Cruz, California
Aug. 2025 – Jun 2026

Indus Institute of Technology and Engineering [3.94 GPA]
Bachelor of Technology in Computer Engineering

Ahmedabad, India
Aug. 2020 – July 2024

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, R

Machine Learning: PyTorch, TensorFlow, Keras, Transformers, scikit-learn, NumPy, Pandas, Matplotlib

ML Systems and Compilers: CUDA, Triton, C++20, pybind11, custom IRs, operator fusion, High-Performance Computing (HPC), kernel selection, memory/execution planning, paged KV caching, quantization

LLM and Data Tools: LangChain, LangGraph, MCP, RAG, Hybrid RAG, ChromaDB, vector databases, Multi-Agent Systems, Agentic AI, Tool Calling, LoRA/QLoRA, LLM Evaluation, LLM Observability, AI Guardrails, Human-in-the-Loop (HITL), PyPDF

Backend and Deployment: Node.js, FastAPI, Flask, REST APIs, FHIR, Docker, CMake, Bazel, Kubernetes, Linux, Git, CI/CD, Nginx

Cloud: AWS EC2, S3, IAM, Lambda, SageMaker, CloudWatch

Data Engineering: PostgreSQL, Redis, Real-Time Data Pipelines, ETL, schema design

Software Engineering: TDD, API integration, DSA, system design

EXPERIENCE

Researcher
[*AI Explainability and Accountability \(AIEA\) Lab @ UCSC*](#)

September 2025 – June 2026
Santa Cruz, California

- Built ReliaGuard Studio and BreakPoint with Next.js, TypeScript, FastAPI, PostgreSQL, Docker, and MLflow to detect over/underreliance in human-AI decisions and compile real LLM failure modes into adversarial eval suites using neuro-symbolic modeling, conformal gating, multi-judge validation, and explainable dashboards.

Artificial Intelligence Engineer
[*Growerx*](#)

August 2024 – August 2025
Ahmedabad, Gujarat, India

- Built a healthcare prior-authorization and clinical workflow agent using FHIR, hybrid RAG, LangGraph, and tool integrations, including early adoption of the newly introduced Model Context Protocol (MCP), with automated evaluations and clinician approval gates to retrieve and validate patient evidence while supporting HIPAA-compliant workflows. Built production multi-agent systems for a retail/e-commerce client integrating CRM, inventory, and customer-support platforms to automate service, product recommendations, and order resolution, with observability, guardrails, and human escalation.

AI Researcher (Part-time)
[*AI Institute of South Carolina*](#)

June 2024 – May 2025
Columbia, South Carolina (remote)

- Conducted research on multimodal toxicity mitigation and anomaly detection, using PostgreSQL-backed experiment tracking to manage dataset metadata, model outputs, evaluation metrics, and benchmarking results for projects like DE-HATE and Time Series Foundation Model evaluation.

Data Scientist
[*Space Applications Centre, ISRO*](#)

January 2024 – May. 2024
Ahmedabad, Gujarat, India

- Built an end-to-end LSTM satellite clock-bias prediction platform with Apache Kafka live streaming and a Next.js/React/TypeScript UI, automating preprocessing and anomaly filtering to improve navigation timing forecasts.

Software Engineer (AI)
[*Inteltytics*](#)

February 2022 – January 2024
Ahmedabad, Gujarat, India

- Coordinated with multiple US-based clients to design and build production agentic AI, LLM, and ML solutions using RAG, tool calling, hybrid retrieval, and LoRA/QLoRA. Developed FastAPI services, integrated enterprise data from Snowflake and Neo4j/GraphRAG, and deployed scalable workflows on Kubernetes with real-time data pipelines and production evaluation/observability.

PROJECTS

CacheIR [CUDA, Triton, C++20, custom compiler IRs, PyTorch, CMake, pybind11, FastAPI, NumPy]

- A narrow, inspectable compiler/runtime for GPU-accelerated decoder-only transformer inference, up to 3.24× faster than vLLM on certain workloads. It imports Llama/Mistral/Qwen models into a custom IR, applies transformer-specific compiler optimizations, plans memory/execution, selects CUDA kernels, and serves through a reference runtime.

BreakPoint [LLM evaluation pipelines, multi-judge validation, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, MLflow, DuckDB]

- A failure-to-eval compiler for LLM systems that mines real failure modes into versioned adversarial regression suites with DSL-defined traps/rubrics, multi-judge calibration, RAG/tool simulators, CI gates, FastAPI/Next.js dashboards, and 360 validated eval cases across 9 failure families.

ReliaGuard Studio [Neuro-symbolic AI, conformal prediction, FastAPI, Next.js, TypeScript, PostgreSQL, MLflow, Docker, scikit-learn]

Research Forge – A Self-Improving Research Agent [LangGraph, autonomous agents, Neo4j, OpenAI API, Pydantic, Streamlit, Python]

Meeting AI Assistant [Real-time multimodal AI, Electron, React, OpenAI API, Node.js, Express, TypeScript]

Time Series Foundation Model Evaluation for Anomaly Detection [TimeGPT, Chronos, Time-MOE, PyTorch, TensorFlow, scikit-learn, Pandas, Python]

Knowledge Graph Generation from Large Language Models [GraphRAG, GPT-4, LLaMA-2, BERT, Transformers, Python]

PUBLICATIONS

Leveraging LSTM for Predictive Modeling of Satellite Clock Bias

- Presented at [CDMA 2025 conference](#) and available on *IEEE Xplore*

Generating Knowledge Graphs from Large Language Models

Time Series Foundational Models: Their Role in Anomaly Detection and Prediction

- Presented at [AAAI 2025 Anomaly Detection in Scientific Domains Workshop](#)

Med-Bot: An AI-Powered Assistant to Provide Accurate and Reliable Medical Information